

IMO 2026 — Problem 6

Solution by DeepSeek V4 Pro

2026-07-21

Source: `results/deepseek-v4-pro/problem-06/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let $a_1, a_2, a_3, \dots$ be an infinite sequence of positive integers greater than 1. Suppose that for all positive integers n , the number a_{n+1} is the smallest positive integer greater than a_n such that $\gcd(a_{n+1}, a_i) > 1$ for every $i = 1, 2, \dots, n$. Prove that there exist positive integers T and L such that $a_{n+T} = a_n + L$ for every positive integer n . (Note that $\gcd(x, y)$ denotes the greatest common divisor of positive integers x and y .)

Solution document

Status

partial — developing rigorous proof

Problem

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Approaches tried

- Computational exploration: sequences appear to be eventually periodic with constant difference; observed two regimes (arithmetic progression when a single prime dominates, or periodic sequence with period >1 governed by a finite “core” set of primes).
- Minimal hitting set / transversal hypergraph approach: each term’s prime set must intersect all previous prime sets; the sequence is exactly the sorted list of numbers whose prime set contains one of the minimal transversals of the infinite family.
- The set of “good” numbers (having $\gcd > 1$ with all terms) is closed under taking multiples; the primitive minimal good numbers correspond to minimal transversals.

- Conjecture: There are only finitely many primitive good numbers; they involve only primes from the first few terms. The sequence is the set of all multiples of these primitive numbers, hence periodic.

Current best

We are constructing a rigorous proof based on the following outline:

1. Let $P(x)$ be the set of prime divisors of x . The condition $\gcd(a_{n+1}, a_i) > 1$ for all $i \leq n$ means $P(a_{n+1}) \cap P(a_i) \neq \emptyset$ for all $i \leq n$.
2. Define the family $\mathcal{F} = \{P(a_n) : n \geq 1\}$. A number $x > 1$ is “good” iff $P(x)$ is a transversal of $\mathcal{F}$ (i.e., $P(x) \cap E \neq \emptyset$ for every $E \in \mathcal{F}$).
3. The sequence (a_n) consists exactly of all good numbers $\geq a_1$ in increasing order. (Because each a_n is good, and if some good number x were missing, the greedy algorithm would have picked it when the sequence passed below it.)
4. Let $\mathcal{T}$ be the family of minimal transversals of $\mathcal{F}$ (with respect to inclusion). Then a number is good iff its prime set contains some member of $\mathcal{T}$.
5. Therefore the good numbers are the union over $H \in \mathcal{T}$ of the multiples of $\prod_{p \in H} p$. If $\mathcal{T}$ is finite, this is a finite union of arithmetic progressions, and the sequence is periodic.
6. **Key lemma:** $\mathcal{T}$ is finite. Moreover, all members of $\mathcal{T}$ consist of primes that appear in the first three terms of the sequence.
 - *Proof:* Let p be the smallest prime divisor of a_1 . Then $a_2 = a_1 + p$, and $P(a_1) \cap P(a_2) = \{p\}$.
 - If all terms are divisible by p , the sequence is arithmetic ($T = 1, L = p$).
 - Otherwise, let $k \geq 3$ be the first index with $p \nmid a_k$. Then $P(a_k)$ contains a minimal transversal H of $\{P(a_1), \dots, P(a_{k-1})\}$ with $p \notin H$.
 - After adding $P(a_k)$, the singleton $\{p\}$ is no longer a transversal. The new minimal transversals are subsets of $\bigcup_{i=1}^k P(a_i)$.
 - One shows that the family $\mathcal{T}$ stabilizes within at most one more step, and that no prime first appearing after a_k can enter any minimal transversal (because such a prime would be redundant—the new term already contains an old minimal transversal hitting all earlier edges).
 - Hence $\mathcal{T}$ is a finite family of subsets of the finite set $\bigcup_{i=1}^{k+1} P(a_i)$.
7. With $\mathcal{T}$ finite, let $M = \text{lcm}\{\prod_{p \in H} p : H \in \mathcal{T}\}$. Then the set of good numbers is a union of residue classes modulo M . Let T be the number of good residues modulo M that are $\geq a_1$ in each block? Actually, since the good numbers are exactly the numbers $\geq a_1$ that are $\equiv r \pmod{M}$ for some residue r in a fixed set R , the sorted list satisfies $a_{n+T} = a_n + M$, where T is the number of good residues in a complete system modulo M . (We need to check that the density is constant from the start; this holds if a_1 itself is the smallest good number, which is true because all smaller numbers are either ≤ 1 or coprime to a_1 ? Actually numbers between 1 and a_1 might be good; but the sequence starts at a_1 , and the periodicity is checked from $n = 1$. We can adjust T and L accordingly.)

We still need to complete the details of step 6 (stabilization of minimal transversals) and step 7 (deriving exact periodicity from the start).

Full proof

(To be written after completing the rigorous arguments.)